3.1.1
Find a solution using the Simplex method
MAX Z = 40x1 + 50x2
subject to
x1 + 2x2 <= 40
4x1 + 3x2 <= 120
and x1,x2 >= 0

Solution:
Problem is
Max $Z = 40x_1 + 50x_2$
subject to
$x_1 + 2x_2 \leq 40$
$4x_1 + 3x_2 \leq 120$
and $x_1, x_2 \geq 0$;

The problem is converted to canonical form by adding slack, surplus, and artificial variables as appropriate
1. As the constraint-1 is of type '≤' we should add slack variable S_1
2. As the constraint-2 is of type '≤' we should add slack variable S_2

After introducing slack variables
Max $Z = 40x_1 + 50x_2 + 0S_1 + 0S_2$
subject to
$x_1 + 2x_2 + S_1 = 40$
$4x_1 + 3x_2 + S_2 = 120$
and $x_1, x_2, S_1, S_2 \geq 0$

Iteration-1

Cj	*Basic Variables*	*Quantity*	40	50	0	0	**Min Ratio** *Quantity/ pivot column*
			x1	***x2***	***S1***	***S2***	
0	***S1***	40	1	**(2)**	1	0	**40/2**=20→
0	*S2*	120	4	3	0	1	120/3=40
	Zj	0	**0**	**0**	**0**	**0**	
	Cj-Zj		40	50↑	0	0	

Positive maximum C_j-Z_j is 50 and its column index is 2. So, the entering variable is x_2.
The minimum ratio is 20 and its row index is 1. So, the leaving basis variable is S_1.

∴ The pivot element is 2.
Entering $=x_2$, Departing $=S_1$, Key Element $=2$

$R_1(\text{new})=R_1(\text{old})\div 2$

$R_1(\text{old})=$	40	1	2	1	0
$R_1(\text{new})=R_1(\text{old})\div 2$	20	0.5	1	0.5	0

$R_2(\text{new})=R_2(\text{old})-3R_1(\text{new})$

$R_2(\text{old})=$	120	4	3	0	1
$R_1(\text{new})=$	20	0.5	1	0.5	0
$3\times R_1(\text{new})=$	60	1.5	3	1.5	0
$R_2(\text{new})=R_2(\text{old})-3R_1(\text{new})$	60	2.5	0	-1.5	1

Iteration-2

Cj	*Basic Variables*	*Quantity*	40	50	0	0	**Min Ratio**
			x1	*x2*	*S1*	*S2*	*Quantity/ pivot column*
50	x_2	20	0.5	1	0.5	0	200/.5=40
0	***S2***	60	**(2.5)**	0	-1.5	1	60/2.5=24→
	Zj	1000	**25**	**50**	**25**	**0**	
	C_j-Z_j		15↑	0	-25	0	

Positive maximum C_j-Z_j is 15 and its column index is 1. So, the entering variable is x_1.
The minimum ratio is 24 and its row index is 2. So, the leaving basis variable is S_2.

∴ The pivot element is 2.5.
Entering $=x_1$, Departing $=S_2$, Key Element $=2.5$

$R_2(\text{new})=R_2(\text{old})\div 2.5$

$R_2(\text{old})=$	60	2.5	0	-1.5	1
$R_2(\text{new})=R_2(\text{old})\div 2.5$	24	1	0	-0.6	0.4

$R_1(\text{new})=R_1(\text{old})-0.5R_2(\text{new})$

$R_1(\text{old})=$	20	0.5	1	0.5	0
$R_2(\text{new})=$	24	1	0	-0.6	0.4
$0.5\times R_2(\text{new})=$	12	0.5	0	-0.3	0.2
$R_1(\text{new})=R_1(\text{old})-0.5R_2(\text{new})$	8	0	1	0.8	-0.2

Iteration-3

C_j	*Basic Variables*	*Quantity*	40	50	0	0	MinRatio
			x_1	x_2	S_1	S_2	
50	x_2	8	0	1	0.8	-0.2	
40	x_1	24	1	0	-0.6	0.4	
	Zj	**1360**	**40**	**50**	**16**	**6**	
	C_j-Z_j		0	0	-16	-6	

Since all C_j-$Z_j \leq 0$

Hence, optimal solution is arrived with value of variables as : $x_1=24, x_2=8$

Max $Z=1360$